KITS

KKR&KSRINSTITUTEOFTECHNOLOGYANDSCIENCES (AUTONOMOUS)
Vinjanampadu, Vatticherukuru Mandal, Guntur, Andhra Pradesh 522017

Programme: B.Tech

Year / Semester: IV/ I
Branch / Section: ECE /I&II&III
Course Code: 23EC7T01
Subject :CELLULAR MOBILE COMMUNICATION

AcademicYear:2026-27

QUESTIONBANKFORMID-1

UNIT-1

Short Answer questions

Q. No.	Question	M	CO	LL
1	State the limitations of conventional mobile telephone systems.	2	C401.1	L1
2	Draw the frequency re use pattern for a cluster size of N=3and N=7.	2	C401.1	L3
3	Explain frequency reuse necessary in cellular networks?	2	C401.1	L1
4	Describe co-channel interference and its impact on cellular system performance..	2	C401.1	L2
5	Discuss the relationship between frequency reuse and co-channel interference	2	C401.1	L2
6	Explain the concept of trunking and its importance in cellular systems.	2	C401.1	L2
7	Define Grade of Service(GOS).	2	C401.1	L1
8	Explain fading and classify its different types.	2	C401.1	L2
9	Explain the significance of coherence bandwidth in wireless communication	2	C401.1	L2
10	What is Doppler spread, and how does it affect mobile communication?	2	C401.1	L1

Essay questions

Q. No.		Question	M	CO	LL
1	a	Explain the limitations of conventional mobile telephone systems.	6	C401.1	L2
	b	Draw and explain the architecture of basic cellular mobile communication system ?	6	C401.1	L3
2	a	Explain the evolution and main features of first,second,third and fourth generation cellular systems.	6	C401.1	L2
	b	Explain the operation of a cellular communication system with frequency reuse and handoff ?	6	C401.1	L2
3	a	Explain the uniqueness of the mobile radio environment.	6	C401.1	L2
	b	Discuss fading ,coherence band width and Doppler spread.	6	C401.1	L2
4	a	Explaintheconceptoffrequencyreuseincellularsystemswithasuitablediagram.	6	C401.1	L2
	b	Determine the number of cells in clusters for the following values of the shift parameters *i* **And *j* in a regular hexagon geometry pattern:** (i) i=2 and j=4 i=3 and j=3.	6	C401.1	L2
5	a	Derive the co-channel interference reduction factor for a cellular system.	6	C401.1	L3
	b	Explain the desired C/I ratio for the normal case in an omni directional antenna system.	6	C401.1	L2
6	a	Explain trunking in cellular systems.	6	C401.1	L2
	b	Calculate the Grade of Service for a given traffic load and number of channels ?	6	C401.1	L3

UNIT-2

Shortanswer questions

Q. No.	Question	M	CO	LL
1	Discuss the effect of co-channel interference on system performance	2	C401.2	L2
2	Distinguish between co-channel and non-co-channel interference.	2	C401.2	L2
3	Define adjacent channel interference.	2	C401.2	L1
4	Explain the causes of near-end interference.	2	C401.2	L2
5	Explain is far-end interference?	2	C401.2	L1
6	List out the types of antennas used at cell site.	2	C401.2	L1
7	How does space diversity reduce fading?	2	C401.2	L1
8	Explain the role of antenna polarization in diversity recept	2	C401.2	L2
9	Explain the relationship between frequency diversity and coherence bandwidth.	2	C401.2	L2
10	Discuss the advantages and limitations of time diversity.	2	C401.2	L2

Essay questions

Q. No.		Question	M	CO	LL
1	a	Describe a method for measuring real-time co-channel interference in a cellular network	6	C401.2	L2
	b	Discuss the methods used to reduce co-channel interference.	6	C401.2	L2
2	a	Derive the expression for carrier-to-interference ratio in a cellular system for normal case and worst-case scenario with an omni-directional antenna.	6	C401.2	L3
	b	Discuss the effects of antenna parameters on cellular system performance.	6	C401.2	L2
3	a	Explain space diversity with a suitable diagram.	6	C401.2	L2
	b	Differentiate between time diversity and frequency diversity	6	C401.2	L3
4	a	Explain time diversity and its significance in mobile communication.	6	C401.2	L2
	b	Compare space ,polarization diversity techniques.	6	C401.2	L3
5	a	Explain adjacent channel interference and its causes.	6	C401.2	L2
	b	Discuss the effects of near-end and far-end interference.	6	C401.2	L2
6	a	Explain the effect of power decrease on cellular coverage and interference.	6	C401.2	L2
	b	Explain the effect of antenna height decrease on coverage and interference.	6	C401.2	L2

UNIT-3

Shortanswer questions

Q. No.	Question	M	CO	LL
1	Discuss the importance of frequency management.	2	C401.3	L2
2	Explain the principle of cell sectorization	2	C401.3	L2
3	Distinguish between channel numbering and channel grouping	2	C401.3	L3
4	Explain the role of an access channel during call establishment?	2	C401.3	L1
5	Define fixed channel assignment.	2	C401.3	L1

Essay questions

Q. No.	Question	M	CO	LL
1	Describe the basic principles of frequency management.	6	C401.3	L2
2	Describe the procedure of channel numbering and grouping	6	C401.3	L2
3	Discuss the importance of paging channels in mobile communication..	6	C401.3	L2
4	Explain channel assignment to cellsites and mobile units.	6	C401.3	L2
5	Differentiate between channel sharing and channel borrowing.	6	C401.3	L3
6	Describe the operation of a sectorized cellular system.	6	C401.3	L2

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